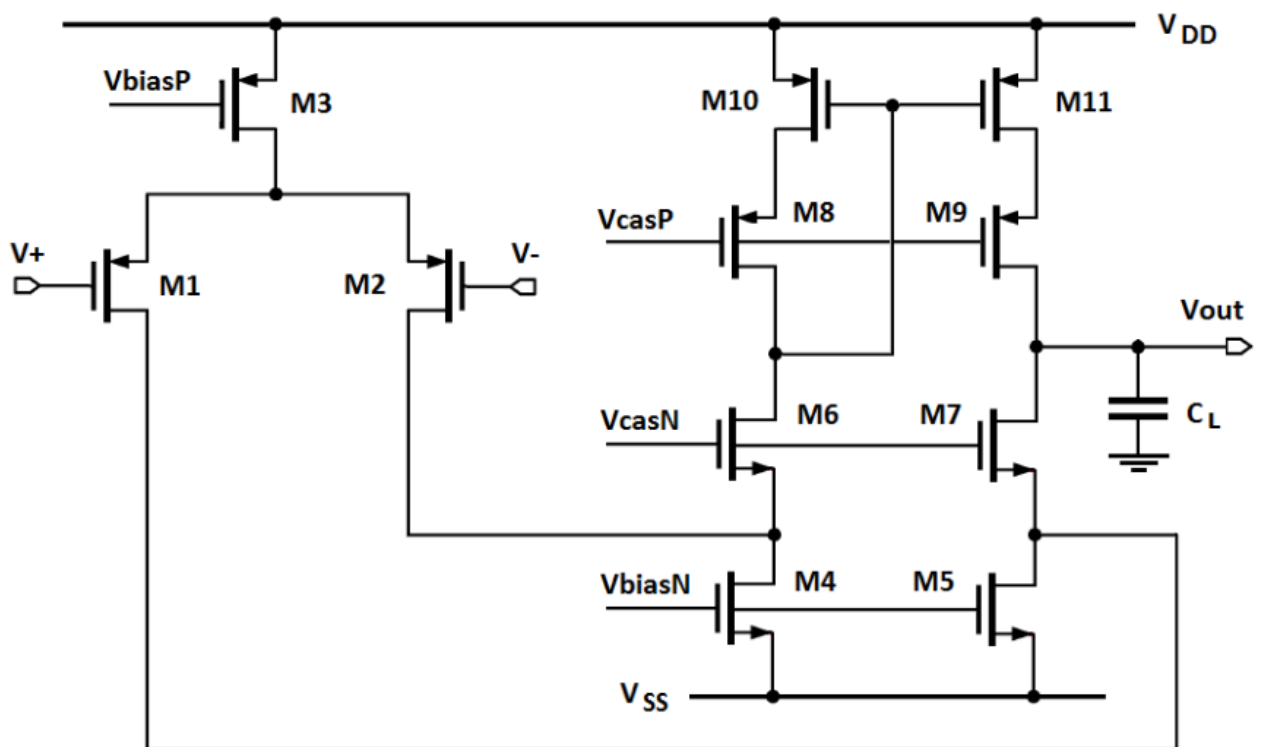


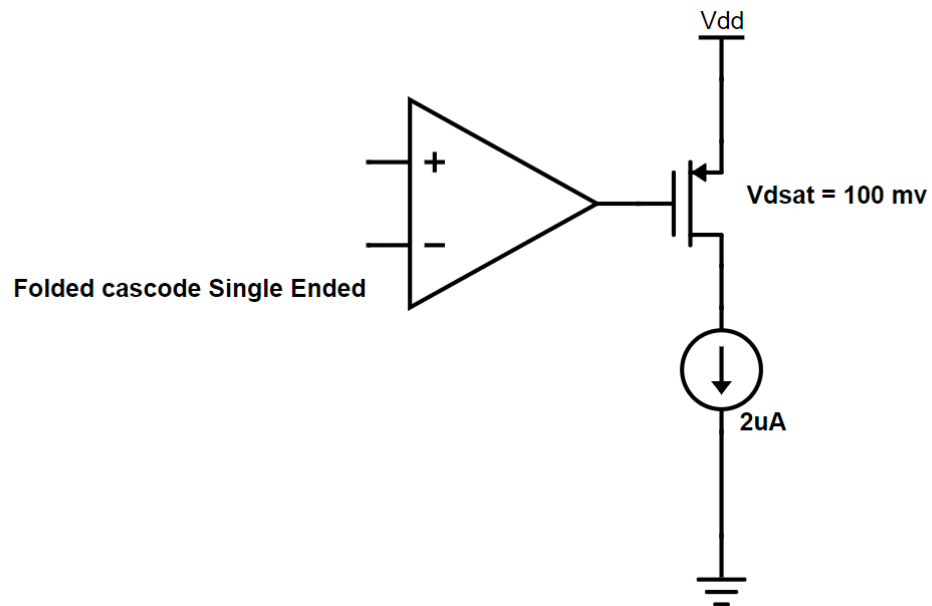
## Lab 10

### Folded Cascode

### Folded Cascode Specs

| Spec                            | Value                               |
|---------------------------------|-------------------------------------|
| VDD                             | 1.2V with 10% variations            |
| Temperature                     | 27 with range (-40 to 125)          |
| Current in OTA                  | 16 $\mu$ A                          |
| Input Common Mode               | 0 to 0.7                            |
| Output                          | Single ended (look the model below) |
| DC Gain                         | 50 dB                               |
| Load cap                        | 1 pF                                |
| GBW                             | 15 MHz                              |
| Input current from VDD          | 1 $\mu$ A                           |
| Biassing network current budget | 5 $\mu$ A                           |





## Simulations

- 1) Report the Schematic of the OTA with DC node voltages clearly annotated with VICM at 0.5V
- 2) Check the whole range of VICM and report the DC results
- 3) Run AC Analysis, report gain and GBW for the whole VICM range
- 4) Run PVT simulations